

FAULT CODE 559 - Injector Metering Rail 1 Pressure - Data Valid But Below Normal Operating Range - Moderately Severe Level

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3823996 - female Weather-Pack™ test lead, and Part Number 3824774 - breakout cable.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Read the fault codes.	Fault Code 451, 452, 2215, 2261, or 2262 active or many inactive counts of Fault Code 451, 452, 2215, 2261, or 2262?
STEP 2.	Clear the operation of the high pressure fuel system.	
	STEP 2A. Check for external leak.	External fuel leaks?
	STEP 2B. Check for air in fuel.	Air present in the fuel flow line?

STEPS		SPECIFICATIONS
	STEP 2C. Check the Stage 1 fuel filter restriction.	Fuel filter restriction less than specifications?
	STEP 2D. Check the Stage 2 fuel filter restriction.	Fuel filter restriction less than specifications?
	STEP 2E. Check the operation of the mechanical dump valve.	Mechanical dump valve leaking?
	STEP 2E. Check the operation of the fuel pump pressurizing assembly.	Fault Code 559 become active?
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault codes.	Fault Code 559 inactive?
	STEP 3A. Clear the fault codes.	All fault codes cleared?

STEP 1. Check the fault codes.

STEP 1A. Read the fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Read the fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 451, 452, 2215, 2261, or 2262 active or many inactive counts of Fault Code 451, 452, 2261, or 2262? YES	Appropriate troubleshooting tree
	Fault Code 451, 452, 2215, 2261, or 2262 active or are there many inactive counts of Fault Code 451, 452, 2261, or 2262? NO	2A

STEP 2. Clear the operation of the high pressure fuel system.

STEP 2A. Check for external leak.

Conditions :

- Operate engine.

Action	Specification/Repair	Next Step
Check for external fuel leaks. • Start the engine and let it idle. • Inspect for external fuel leaks.	External fuel leaks? YES Repair : Repair all fuel leaks. Reference the QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 Modular Common Rail System Service Manual, Bulletin 4021592.	3A
	External fuel leaks? NO	2B

STEP 2B. Check for air in fuel.

Conditions :

- Remove air bleed line from air bleed valve on the fuel drain manifold block.
- Route the air bleed line into a suitable container to collect fuel.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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Inspect the fuel flow for air. Use the following procedure in Service Manual, K38, K50, QSK38 and QSK50, Bulletin 4021528. Refer to Procedure 006-003 in Section 6 (/qs3/pubsys2/xml/en/procedures/28/28-006-003.html) Use the following procedure in Service Manual, QSK45 and QSK60, Bulletin 4021530. Refer to Procedure 006-003 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-003.html)	Air present in the fuel flow line? YES Repair : Repair or replace the damaged line or loose connection. Refer to Procedure 006-024 in the Associated Procedures Table.	3A
	Air present in the fuel flow line? NO	2C

STEP 2C. Check the Stage 1 fuel filter restriction.

Conditions :

- Turn keyswitch OFF.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Measure the fuel restriction. • Operate the engine at idle and measure the restriction across the first stage fuel filters. • Refer to Procedure 006-020 in Section 6.	Fuel filter restriction less than specifications? YES	2D
	Fuel filter restriction less than specifications? NO Repair : The restriction across the first stage filter is above specification. Replace the first stage fuel filter. Refer to Procedure 006-075 in the Associated Procedures Table.	3A

STEP 2D. Check the Stage 2 fuel filter restriction.

Conditions :

- Turn keyswitch OFF.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Measure the fuel restriction. • Operate the engine at high idle and measure the restriction across the second stage fuel filters. • Refer to Procedure 006-020 in Section 6.	Fuel filter restriction less than specifications? YES	2E
	Fuel filter restriction less than specifications? NO Repair : The restriction across the second stage filter is above specification. Replace the second stage fuel filter. Refer to Procedure 006-076 in the Associated Procedures Table.	3A

STEP 2E. Check the operation of the mechanical dump valve.

Conditions :

- Disconnect the mechanical dump valve drain hose from the fuel drain manifold.
- Place the relief drain hose into a clean container.
- Operate the engine.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Start the engine. • Use INSITE™ electronic service tool High Pressure Common Rail Pressure Test to raise the injector metering rail 1 pressure. • Measure the fuel leakage from the mechanical dump valve. • Install and M18 plug into the fuel drain manifold. If the engine will not start, reference the Engine Performance Troubleshooting Tree in Section TT.	Mechanical dump valve leaking? YES Repair : Replace the mechanical dump valve and fuel pump pressurizing assembly. Refer to Procedure 006-061 in the Associated Procedures Table. Refer to Procedure 005-016 in the Associated Procedures Table.	2F
	Mechanical dump valve leaking? NO Repair : Replace the fuel pump pressurizing assembly. Refer to Procedure 005-016 in the Associated Procedures Table.	2F

STEP 2F. Check the operation of the fuel pump pressurizing assembly.

Conditions :

- Turn keyswitch ON.
- Operate the engine.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Use INSITE™ electronic service tool to verify the fault codes. - Start the engine and let it idle for 1 minute. - Use INSITE™ electronic service tool High Pressure Common Rail Pressure Test to raise the injector metering rail 1 pressure.	Fault Code 559 become active? YES Repair : Replace the high pressure fuel pump assembly. Refer to Procedure 005-016 in the Associated Procedures Table.	3A
	Fault Code 559 become active? NO	3A

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault codes.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault codes. - Start the engine and idle for 1 minute. - Use INSITE™ electronic service tool to verify the inactive fault codes.	Fault Code 559 inactive? YES	3B
	Fault Code 559 inactive? NO	1A

STEP 3B. Clear the fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO	Appropriate troubleshooting steps

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Fuel Pump	Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/56/56-005-016-tr.html)	QSK45 and QSK60	4021530
Fuel Inlet Restriction	Refer to Procedure 006-020 (/qs3/pubsys2/xml/en/procedures/56/56-006-020-tr.html)	QSK45 and QSK60	4021530
Fuel Supply Lines	Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/56/56-006-024-tr.html)	QSK45 and QSK60	4021530
Fuel Pressure Relief Valve	Refer to Procedure 006-061 (/qs3/pubsys2/xml/en/procedures/102/102-006-061.html)	QSK45 and QSK60	4021530
Fuel Filter (Stage 1)	Refer to Procedure 006-075 (/qs3/pubsys2/xml/en/procedures/56/56-006-075-tr.html)	QSK45 and QSK60	4021530
Fuel Filter (Stage 2)	Refer to Procedure 006-076 (/qs3/pubsys2/xml/en/procedures/56/56-006-076-tr.html)	QSK45 and QSK60	4021530

